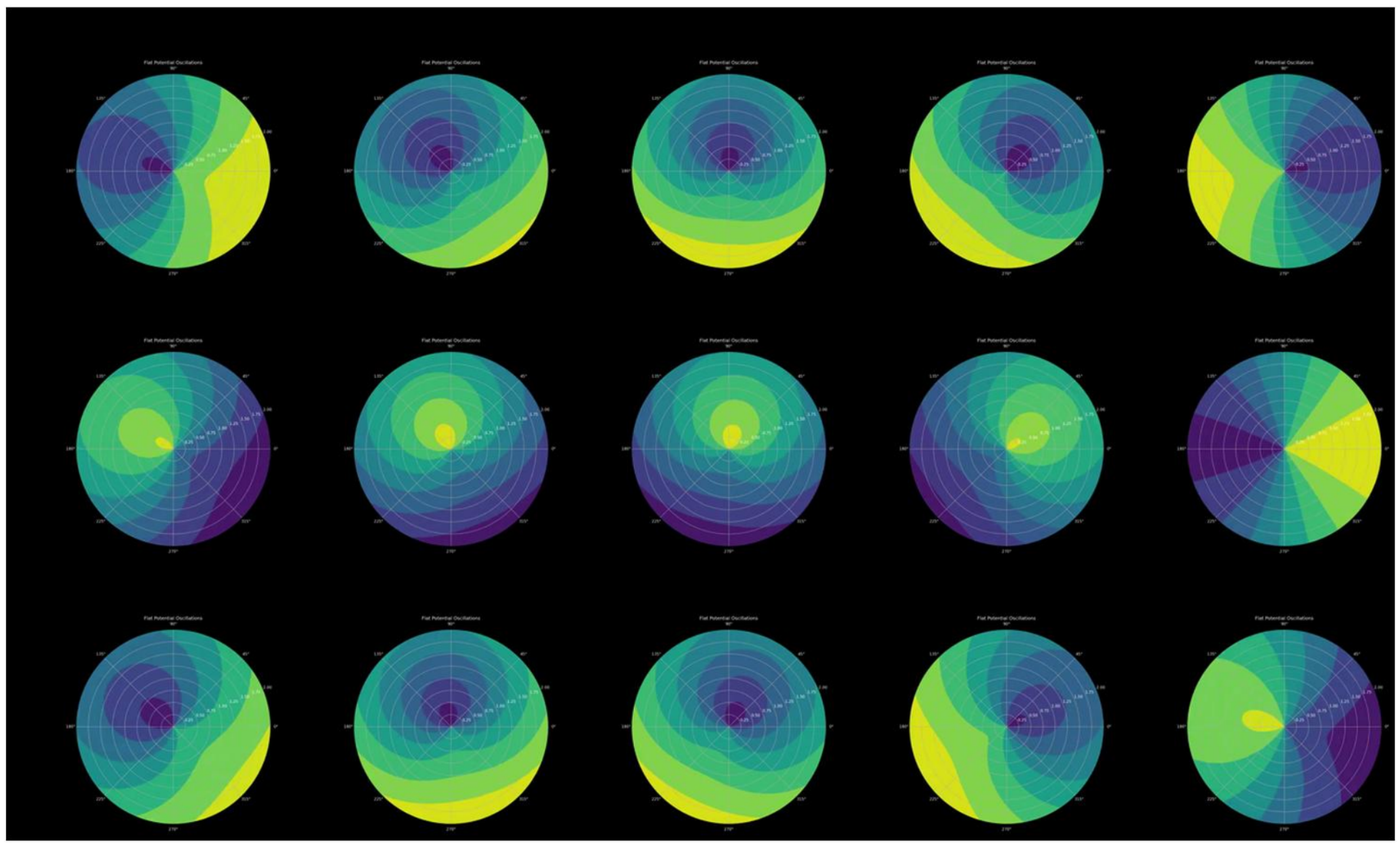
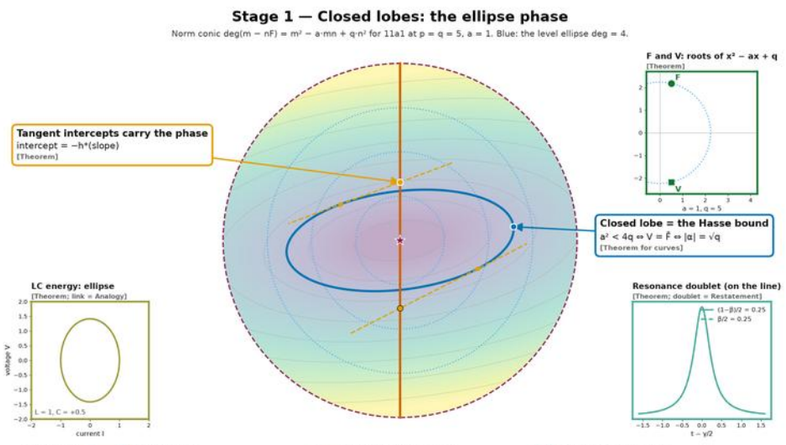
The animation, labelled: what each element is in mathematics (RH Atlas v1.1)

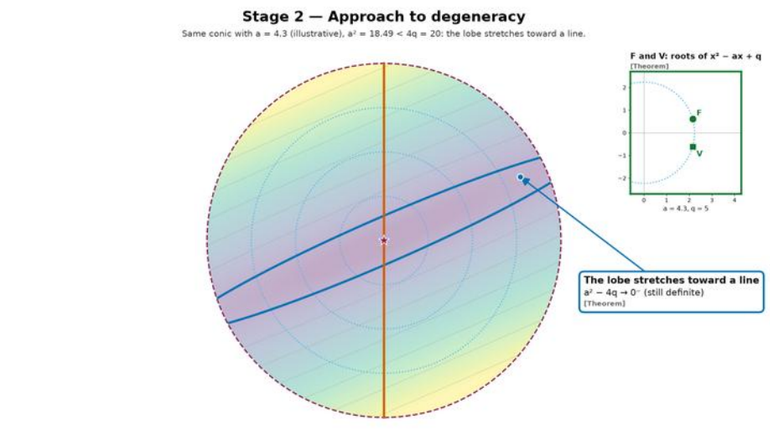
One colour per concept throughout; grey [badges] give the status. Re-rendered by Grok Bot; not J. W. McGreevy's code. RH is not proved.

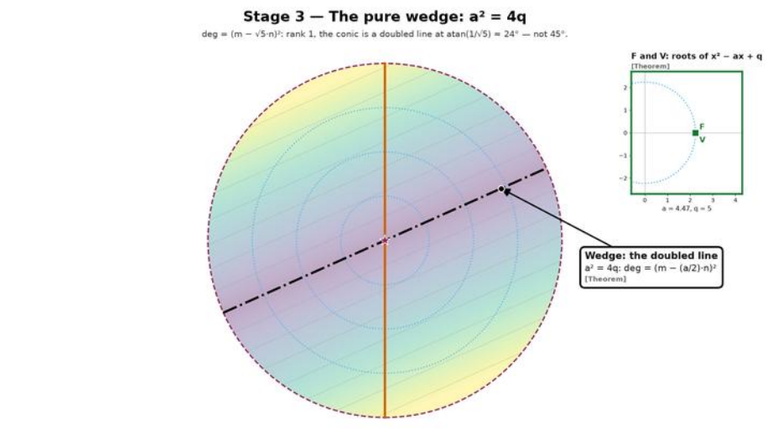
1 The original: J. W. McGreevy, "Flat Potential Oscillations" (Aug 2025) — x.com/McGreevyMD/status/1951474097990279331



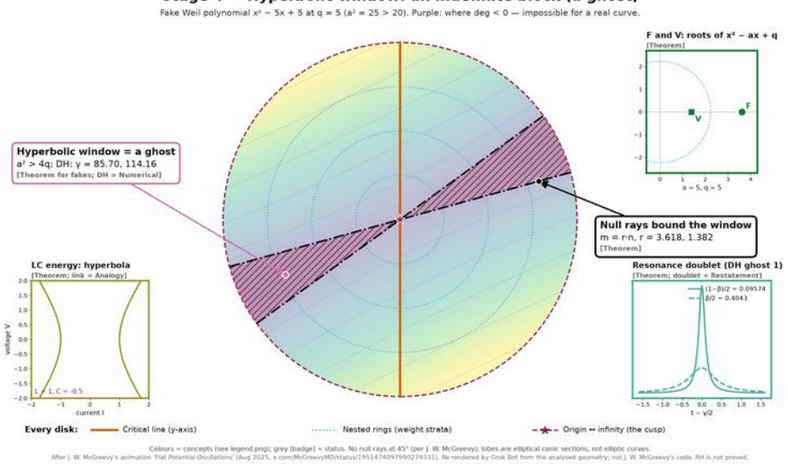
2 Stages 1–3: closed lobe → approach → pure wedge

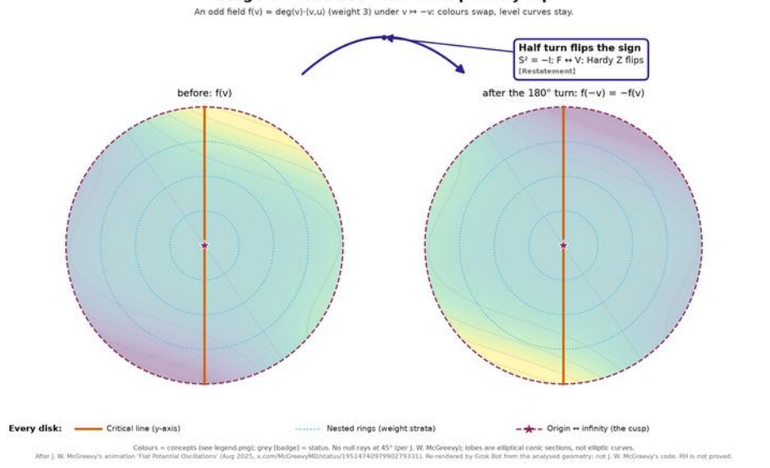


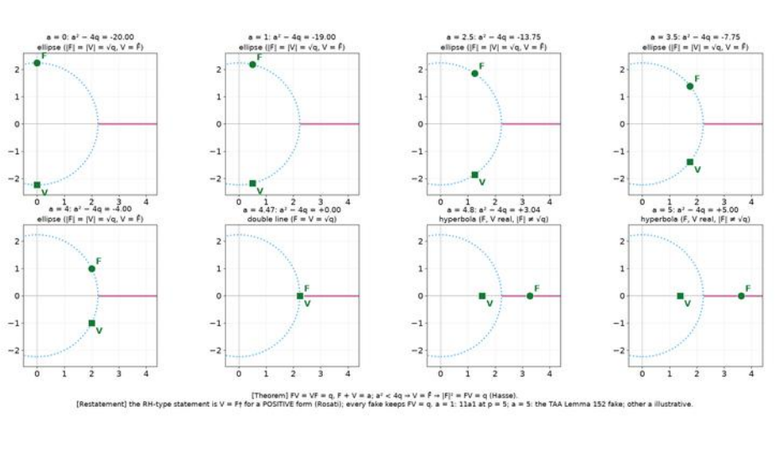




3 Stages 4–5 and the F ↔ V sweep: ghost window → half turn → roots of x² − ax + q







4 Legend: one colour per concept (status = grey badge)

Legend — one colour per concept

How to read this. Each colour is ONE concept, used for the same thing in every frame, the overview and the GIF: the drawn element (line, dots, outline, region, inset) and the box that names it share that colour. The grey [badge] in each box is the status: Theorem = proved; Restatement = equivalent, adds no positivity; Analogy = a parallel, not a proof step; Numerical = computed in a TAA lemma. The second cue (line style or marker) is shown in the swatch. RH is not proved.

Concept	In plain words	Math	Status · refs
Critical line (y-axis) #D55E00	Where every zero should sit; scaling is unitary only here	Re s = ½: |λ^(½−s)| = 1 for all λ	[Theorem; axis = Analogy] TAA Lemma 153 (critical line as the spectrum of unitary scaling)
Nested rings (weight strata) #56B4E9	Each ring is one decay rate; purity means everything sits on one ring	|z| = q^(−σ); purity = one ring	[Restatement for Frobenius; Analogy for ζ] TAA Lemma 153 (distortionless cusp: purity ↔ Heaviside condition)
Origin ↔ infinity (the cusp) #882255	The centre and the boundary are the two ends of the cusp, where waves enter and leave	$a_0(y) = y^s + \varphi(s)\cdot y^{(1-s)}$, |φ| = 1 on the line	[Analogy for the identification] TAA Lemma 153 (cusp: ||φ(½+it)| − 1| ≤ 3.9·10^(−31) for t ≤ 60)
Tangent intercepts (Legendre) #E69F00	Tangent lines meet the y-axis at the Legendre transform: they carry the phase	intercept = −h*(slope); amplitude |S″|^(−½)	[Theorem] TAA Lemma 153 (Legendre ↔ Fourier: stationary phase, rel. error 0.003 at λ = 80)
Closed lobe (ellipse) = Hasse #0072B2	The degree form is positive, so its level sets are ellipses; that is the Hasse bound	a² < 4q ⇔ V = F̄ ⇔ |α| = √q	[Theorem for curves] TAA Lemma 152 (norm conic); TAA Lemma 153 (11a1: max|a_p|/√p = 1.947, 302 primes)
Wedge: double line / null cone #000000	At the edge the ellipse collapses to a doubled line; beyond it, two null lines (wave characteristics)	a² = 4q: deg = (m − (a/2)·n)²	[Theorem] TAA Lemma 152 (TAA Lemma 152, correction 10); TAA Lemma 153 (dual = vertex counted twice)
Hyperbolic window (ghost) #CC79A7	The form takes negative values: an indefinite block, i.e. an off-line zero	a² > 4q: deg < 0 somewhere	[Theorem for fakes; DH = Numerical] TAA Lemma 152 (fake x² − 5x + 5); TAA Lemma 153 (DH windows [84.36,84.77], [85.48,85.91], [114.04,114.29])
Half turn (180°) #332288	Turning by 180° flips the sign of odd data and swaps Frobenius with Verschiebung	S² = −I; F ↔ V; Hardy Z flips sign	[Restatement] TAA Lemma 152 (ζ 38/38 sign changes; DH 62 vs 66 on [10,120]); TAA Lemma 115
Frobenius F and Verschiebung V #117733	The two roots of x² − a·x + q; conjugate on the circle |z| = √q, real and split for a fake	F·V = q, F + V = a	[Theorem; V = F† (positive form) = Restatement] TAA Lemma 115; TAA Lemma 117; TAA Lemma 121
Resonance doublet #44AA99	At the cusp each zero shows as two Breit–Wigner lines; equal widths exactly on the line	widths β/2, (1−β)/2 (sum ½)	[Theorem; doublet = Restatement] TAA Lemma 153 Thm 7.7; TAA Lemma 154 (DH ghost 1: 0.0957 / 0.4042)
LC circuit energy #999933	A circuit stores energy on an ellipse when L, C > 0; a negative C gives a runaway hyperbola	½LI² + ½CV² = E	[Theorem for the conic; Analogy for the link] TAA Lemma 153 (TAA Lemma 153, correction 30: Wick rotation is a different sign flip from C → −C)

Re-rendered by Grok Bot (TAA Lemma 152–154 geometry); original: x.com/McGreevyMD/status/1951474097990279331.
a = 1 is 11a1 at p = 5; a = 5 is the TAA Lemma 152 fake; a = 4.3 is illustrative.